

Invariant Estimator Framework for Autonomous Cars

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I. STATE ON $SE_2(3)$

We define the state on the group manifold $SE_2(3)$, such that we can maintain invariant observer properties while keeping the system as observable as possible:

$$X \triangleq \begin{bmatrix} {}^{\mathcal{W}}R_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} & {}^{\mathcal{W}}\mathbf{p}_{\mathcal{B}} \\ \mathbf{0}_{1 \times 3} & 1 & 0 \\ \mathbf{0}_{1 \times 3} & 0 & 1 \end{bmatrix} \in SE_2(3) \quad (1)$$

$$\boldsymbol{\theta} = (\mathbf{b}_g, \mathbf{b}_a) \in \mathbb{R}^6$$

With the associated tangent vector defined as the following:

$$\boldsymbol{\xi} = [\boldsymbol{\xi}_{\phi} \quad \boldsymbol{\xi}_{\mathbf{v}} \quad \boldsymbol{\xi}_{\mathbf{p}} \quad \boldsymbol{\zeta}_g \quad \boldsymbol{\zeta}_a] \in \mathbb{R}^{15} \quad (2)$$

where the exponential on the state component of the tangent vector defined as $\exp([\boldsymbol{\xi}_{1:9}]^\wedge) = X^{-1} \hat{X}$, and $\boldsymbol{\zeta} = \hat{\boldsymbol{\theta}} - \boldsymbol{\theta}$.

II. PROCESS MODEL

We define the following continuous time (CT) dynamics:

$$\begin{aligned} {}^{\mathcal{W}}\dot{R}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\bar{\boldsymbol{\omega}}^\wedge \\ {}^{\mathcal{W}}\dot{\mathbf{v}}_{\mathcal{B}} &= {}^{\mathcal{W}}R_{\mathcal{B}} {}^{\mathcal{B}}\bar{\mathbf{a}} + {}^{\mathcal{W}}\mathbf{g} \\ {}^{\mathcal{W}}\dot{\mathbf{p}}_{\mathcal{B}} &= {}^{\mathcal{W}}\mathbf{v}_{\mathcal{B}} \\ \dot{\mathbf{b}}_g &= \mathbf{w}_{bg}, \mathbf{w}_{bg} \sim \mathcal{N}(0, \Sigma_g) \\ \dot{\mathbf{b}}_a &= \mathbf{w}_{ba}, \mathbf{w}_{ba} \sim \mathcal{N}(0, \Sigma_a) \end{aligned} \quad (3)$$

where ${}^{\mathcal{B}}\bar{\boldsymbol{\omega}}$ and ${}^{\mathcal{B}}\bar{\mathbf{a}}$ are the bias corrected inputs, which are later derived in ???. And we know that these dynamics, when bias-free, are *group-affine* - that is, that they yield a log-linear error such that $\dot{\boldsymbol{\eta}} = A\boldsymbol{\eta}$ - this is further shown in the appendix. However, this is generally not the case, as shown in (CITE Barreau here), that the augmentation of biases onto the InEKF state yields an imperfect InEKF, as $\bar{\boldsymbol{\omega}}$ and $\bar{\mathbf{a}}$ both depend on $\hat{\mathbf{b}}$.

If we assume that the input measurements ${}^{\mathcal{B}}\boldsymbol{\omega}$ and ${}^{\mathcal{B}}\mathbf{a}$ are constant over some small timestep interval $[t, t+\Delta t]$ (i.e., zero-order hold), then we know that the rotation matrix propagation is defined exactly:

$${}^{\mathcal{W}}R_{\mathcal{B}}(t_k + s) = {}^{\mathcal{W}}R_{\mathcal{B},k} \exp(s[\boldsymbol{\omega}]^\wedge) \quad (4)$$

This means that we can substitute this into the discretization of both velocity and position, with $s = u\Delta t$, for some timestep Δt . For instance, with velocity, the integrand yields:

$$\int_0^{\Delta t} \exp(s[\boldsymbol{\omega}]^\wedge) ds = \Delta t \int_0^1 \exp(u[\boldsymbol{\omega}]^\wedge) du = \Delta t \Gamma_1(\boldsymbol{\omega}) \quad (5)$$

And the integral behaves similarly for position. This means that after discretizing, we are left with the following dynamics that obey the $SO(3)$ manifold that live on $SE_2(3)$, with $\boldsymbol{\phi} \triangleq \bar{\boldsymbol{\omega}}\Delta t$:

$$\begin{aligned} R_{k+1|k} &= R_{k|k} \Gamma_0(\boldsymbol{\phi}) \\ \mathbf{v}_{k+1|k} &= \mathbf{v}_{k|k} + {}^{\mathcal{W}}\mathbf{g}\Delta t + R_{k|k} \Gamma_1(\boldsymbol{\phi}) \mathbf{a} \Delta t \\ \mathbf{p}_{k+1|k} &= \mathbf{p}_{k|k} + \mathbf{v}_{k|k} \Delta t + \frac{1}{2} {}^{\mathcal{W}}\mathbf{g} \Delta t^2 + R_{k|k} \Gamma_2(\boldsymbol{\phi}) \mathbf{a} \Delta t^2 \\ \hat{\boldsymbol{\theta}}_{k+1|k} &= \hat{\boldsymbol{\theta}}_{k|k} \end{aligned} \quad (6)$$